Product Requirement Document (PRD)

Product Name: Secure Team Password Manager & Auto-Fill Extension
Target Domain: Cybersecurity / Team & Personal Productivity
Document Status: Draft

1. Executive Summary

This platform is a zero-knowledge password management solution consisting of a **Web Application** and a **Chrome Extension (Manifest V3)**. It enables users to securely store hundreds of website credentials, organize them, and share access with friends, family, or team members without compromising security.
The system utilizes **Zero-Knowledge End-to-End Encryption (E2EE)**, ensuring that raw passwords are encrypted on the client side before touching the database (e.g., Supabase).

2. Technical Security Architecture

To guarantee privacy, the system must adhere to a **Zero-Knowledge Architecture**:

- **Key Derivation:** Master Password passes through **Argon2id** or **PBKDF2** locally to generate a Symmetric Key (K_{sym}). The Master Password is *never* sent to the server.
- **Asymmetric Key Pair for Sharing:** Every user receives an RSA-2048 or X25519 key pair (Public Key stored on server, Private Key encrypted using K_{sym}).
- **Sharing Protocol:** To share a password with User B:

1. User A decrypts the target password locally.
2. User A encrypts the password using User B's **Public Key**.
3. The re-encrypted payload is sent to the database.
4. User B decrypts it using their own **Private Key**.

3. Web Application Requirements

3.1 Vault Management

- **Credential Storage:** Ability to store Title, Username/Email, Encrypted Password, Web Address (URL), Category (Work, Personal, Social), and Secure Notes.
- **Bulk Import/Export:** Support CSV/JSON import from standard password managers (Bitwarden, 1Password, Chrome CSV).
- **Search & Filtering:** Real-time search across hundreds of credentials by title, domain, or category.

3.2 Sharing & Permissions System

- **Direct User Sharing:** Share individual credentials with another user via email address.
- **Permission Levels:**

Permission Role	View Password	Auto-Fill	Edit Credential	Re-Share
Viewer	Yes	Yes	No	No
Editor	Yes	Yes	Yes	No
Admin / Owner	Yes	Yes	Yes	Yes

- **Team Workspaces / Collections:** Create shared folders where access is granted to an entire group (e.g., "Marketing Team", "Family").

4. Chrome Extension Requirements (Manifest V3)

4.1 Auto-Detection & Capture

- **Login Detection:** Automatically detect when a user submits a login form on a web page.
- **Save Prompt:** Display a floating prompt asking: *"Save login credentials to your Vault?"*
- **Update Prompt:** If the username exists but the password changed, prompt to update the existing vault item.

4.2 Auto-Fill Engine

- **Domain Matching:** Match active browser tab domains (e.g., app.github.com $\rightarrow$ github.com) against vault URLs.
- **Inline Field Injection:** Display a custom icon inside <input type="text"> and <input type="password"> fields.
- **Shared Passwords Auto-Fill:** Shared vault items must seamlessly populate inline options just like personal passwords.
- **Quick Copy Overlay:** Allow users to copy TOTP codes, usernames, or passwords directly from the extension popup.

4.3 Extension UI/UX

- **Quick Search Popup:** Global search keyboard shortcut (e.g., Cmd/Ctrl + Shift + L) to launch extension search and fill instantly.
- **Locking Mechanism:** Auto-lock extension after X minutes of inactivity (requiring Master Password or PIN to unlock).